

DEEPAK REDDY YADAMA

Jacksonville, FL | 516-315-9011 | deepakreddyyadama@gmail.com | [linkedin.com/in/deepak-reddy-7a5944250/](https://www.linkedin.com/in/deepak-reddy-7a5944250/)

PROFESSIONAL SUMMARY

M.S. Computer and Information Science candidate with hands-on experience building Python and SQL data workflows, evaluation pipelines, REST-integrated services, and enterprise-style AI applications. Experience includes RAG pipelines, vector retrieval, model evaluation, data quality checks, experiment tracking, and security-conscious cloud workflows. Well aligned to cyber data engineering work involving governed data foundations, audit-ready evidence, privacy-aware design, and AI-enabled risk and compliance operations.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, C++, C#, HTML, CSS

Data Engineering: ETL/ELT workflows, batch processing, data modeling, data quality checks, dataset curation, metadata-aware workflows, REST API integrations

AI and LLM: RAG, LangChain, LangGraph, AutoGen, prompt engineering, vector retrieval, LLM evaluation, NLP, transformer models

Data and MLOps: PyTorch, TensorFlow, Scikit-learn, MLflow, experiment tracking, model monitoring, CI/CD, automated testing

Platforms: AWS (EC2, S3, IAM, SageMaker), Azure, GCP, Docker, Kubernetes, Git/GitHub

Governance and Security: data governance, privacy-conscious workflows, explainability, bias awareness, access control fundamentals, NIST AI RMF

EDUCATION

University of North Florida - M.S. in Computer and Information Science | Expected May 2026

Malla Reddy Institute of Engineering and Technology - B.Tech. in Computer Science Engineering | 2020 - 2024

EXPERIENCE

Graduate Research Assistant - Agentic AI and Computer Vision Systems | University of North Florida | Aug 2025 - Present

- Designed Python and PyTorch pipelines for data preprocessing, model evaluation, retrieval workflows, and reproducible experiment tracking across AI research projects.
- Built RAG-enabled workflows using vector databases and prompt-based retrieval to generate grounded outputs and compare model responses against supporting evidence.
- Prepared evaluation-ready datasets, quality checks, and reporting workflows for benchmarking, error analysis, and repeatable result generation across research use cases.
- Enhanced SAM-2 segmentation experiments with LoRA adapters, multi-prompt guidance, and depth-aware context features, improving IoU by 0.12 and achieving a Dice score of 0.88.

Simulation Engineer Intern | Tap4 Technologies | May 2025 - Aug 2025

- Developed backend modules in Python and integrated REST APIs to support scalable simulation workflows and AI-assisted data processing services.
- Created monitoring-oriented workflow components to track model behavior, latency, and reporting quality, contributing to a 35% improvement in reporting efficiency.
- Supported reliable engineering delivery through Git-based version control, automated testing, CI/CD practices, and Agile collaboration.
- Applied security-conscious data handling and structured workflow validation to improve system integrity across cloud-based environments.

PROJECTS

Multi-Agent RAG System with LLM Evaluation

- Built a multi-agent document analysis system using LangChain and AutoGen with retrieval, reasoning, and synthesis agents for grounded question answering.
- Implemented a RAG pipeline with FAISS, semantic chunking, evaluation datasets, and answer-quality checks, reaching 91% retrieval precision.
- Designed reusable workflow components for retrieval, prompting, and result comparison to support repeatable LLM experimentation.

Medical Image Classification Service (PathMNIST)

- Built an end-to-end PyTorch pipeline for a 9-class pathology image classification task on the 107K-image PathMNIST dataset covering preprocessing, training, evaluation, and visualization.
- Applied transfer learning with ResNet-18, experiment tracking, and automated evaluation to achieve 86.1% test accuracy and 0.84 macro F1-score.

CERTIFICATIONS

AWS Solutions Architect (Forage)

Azure AI Fundamentals

Cisco Python Essentials

NIST AI RMF Self-Study